

EN4730 Convex Engineering Design Tutorial 1

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1. Consider the following inequalities:

$$\begin{cases} x_1 + x_2 \leq 4 \\ x_1^2 \geq 2 \\ x_2 \geq 0 \end{cases}$$

- (a) Draw the feasible region defined by the above constraints.
- (b) Determine whether the feasible region is a convex set. Justify your answer.

2. Consider the following constraints:

$$\begin{cases} x + y \geq 3, \\ y - 2x \leq 3, \\ y \geq x^2 - 2. \end{cases}$$

where $x, y \in \mathbb{R}$.

- (a) Plot the feasible region defined by these constraints in the (x, y) plane.
- (b) Is the feasible region a convex set? Justify your answer.

3. Verify the following functions are convex or concave or neither convex or concave.

(a) $f(x) = x_1^4 + 6x_1^2x_2^2 + x_2^4$

(b) $f(\mathbf{x}) = -x_1 \log x_1 - x_2 \log x_2$ with $x_1, x_2 \geq 0$

4. Consider the following optimization problem

$$\begin{aligned} & \underset{\mathbf{x}}{\text{minimize}} && f_0(\mathbf{x}) \\ & \text{subject to} && 3x_1 + x_2 \geq 1, \\ & && x_1 + 5x_2 \geq 1, \\ & && x_1 \geq 0, \\ & && x_2 \geq 0. \end{aligned}$$

- (a) Draw the feasible region.
- (b) Is the feasible region a convex set? Justify your answer.

- (c) For each of the following objective functions, find the optimal set and the optimal value.
- i. $f_0(x_1, x_2) = x_1 + x_2$.
 - ii. $f_0(x_1, x_2) = -x_1 - x_2$.
 - iii. $f_0(x_1, x_2) = x_1$.
 - iv. $f_0(x_1, x_2) = \max\{x_1, x_2\}$.
5. It is given that $f_1(\mathbf{x})$ and $f_2(\mathbf{x})$ are convex is convex on the set R_c . Show that $f(\mathbf{x}) = af_1(\mathbf{x}) + bf_2(\mathbf{x})$, where $a, b \geq 0$ is convex.
6. Show that a function $f(\mathbf{x}) \in C^2$ is convex over a convex set R_c if and only if the Hessian $H(\mathbf{x})$ of $f(\mathbf{x})$ is positive semidefinite for all $\mathbf{x} \in R_c$. Here, C^2 class of twice continuously differentiable functions.
 Hint: Use Taylor expansion $f(\mathbf{x} + \boldsymbol{\delta}) = f(\mathbf{x}) + \nabla(\mathbf{x})^T \boldsymbol{\delta} + \frac{1}{2} \boldsymbol{\delta}^T H(\mathbf{x} + \alpha \boldsymbol{\delta}) \boldsymbol{\delta}$ where $0 \leq \alpha \leq 1$ and $\boldsymbol{\delta}$ is a small perturbation vector.

References

- [1] Stephen Boyd, “Convex optimization,” *Cambridge UP*, 2004.
- [2] Andreas Antoniou and Wu-Sheng Lu, *Practical optimization: algorithms and engineering applications*, vol. 19, Springer, 2007.